VISHAL KUMAR

EMBEDDED SYSTEMS ENGINEERING | EMBEDDED C | ESP32 | ARDUINO | IoT

+91 7456817715 | Modinagar, Ghaziabad | vishalkumar211024@gmail.com
linkedin.com/in/vishal-kumar-ba5b16330 | github.com/vishalkumar211024-ux

PROFESSIONAL SUMMARY

Electronics & Communication Engineering student (2027) with hands-on experience in embedded systems, sensor interfacing, microcontrollers, digital electronics, IoT and telemetry simulation. Skilled in C, Python, ESP32, ESP8266 and Arduino, with working knowledge of UART, I2C, SPI and CAN concepts. Experience includes sensor-based automation, GSM alerts, space/satellite monitoring concepts and scenario-based MQTT telemetry, with exposure to testing, Git/GitHub and API development.

EDUCATION

Raj Kumar Goel Institute of Technology, Ghaziabad | B.Tech, Electronics & Communication Engineering | Expected 2027

TECHNICAL SKILLS

Programming: C, Embedded C fundamentals, Python **Microcontrollers:** ESP32, ESP8266, Arduino Uno; basic STM32/ARM
Embedded: GPIO, ADC, sensors, actuators, motor control, firmware fundamentals **Communication:** UART, I2C, SPI, CAN, HC-05, GSM, MQTT
Tools: Arduino IDE, VS Code, Git/GitHub, FastAPI, Uvicorn, Streamlit, simulation tools

INTERNSHIP EXPERIENCE

Space & Satellite Technology Intern / Domain Captain — Uptoskills | 2 Jul 2026 – 2 Oct 2026
- Completed an internship in Space & Satellite Technology and served as domain Captain, contributing to task ownership and team coordination.

Embedded Systems & IoT Intern — Maincrafts Technology
- Worked on embedded/IoT-oriented tasks involving microcontrollers, sensors and data collection; developed understanding of hardware–software integration.

Digital Electronics & VLSI Intern — Codec Technologies Pvt. Ltd. | Jul 2026 – Aug 2026
- Studied digital electronics and VLSI fundamentals, with focus on circuit design and analysis.

SELECTED PROJECTS

Satellite / PV Power Monitoring & Beam Alignment | ESP32, Sensors, CSV Logging, Dashboard
- Worked on control, monitoring and data-logging design for a PV/satellite power-transfer prototype; defined timestamp, voltage, current, power, received power, distance, angle and load-resistance fields.
- Planned sensor integration, dashboard parameters and beam-alignment logic to maximize received power.

MQTT Telemetry Simulator | Python, Pydantic, FastAPI, Pytest, JSON, Git/GitHub
- Developed scenario-based battery telemetry for voltage, current, temperature, SOC and SOH; added validation, sequential battery IDs, JSON export, argparse and automated tests.
- Built FastAPI event-generation and health-check endpoints, including normal and anomaly scenarios.

Smart Energy Meter with GSM Alert | ESP32, GSM, I2C LCD
- Measured voltage/current, calculated power, energy and estimated bill; implemented overload indication and GSM SMS alert logic.

IoT Home Automation | ESP32, DHT22, PIR, LDR
- Designed sensor-based fan and lighting control using temperature, motion and ambient-light readings.

Smart Parking System | ESP8266, IR Sensors, Servo, LCD
- Detected parking-slot availability, displayed status on LCD and automated the entry gate.

Voice Controlled Wheelchair | Arduino Uno, HC-05, L298N
- Implemented Bluetooth voice-command control and motor-driver logic for movement and turning.

CERTIFICATIONS

Space & Satellite Technology — Internship Completion Certificate (Uptoskills, 2 Jul–2 Oct 2026) • Embedded Systems Fundamentals • Arduino & IoT Basics • Digital Electronics & VLSI • Embedded Systems & IoT Internship — Maincrafts Technology

TARGET ROLE: EMBEDDED SYSTEMS ENGINEERING INTERN